Consulting & Analytics Club
IIT Guwahati

Retail Consumer Behavior & Loyalty Analytics

A comprehensive data-driven study of customer segments, purchase behavior, and strategic growth opportunities

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Problem Statement

A leading omnichannel retail company possesses extensive transaction data but lacks a structured analytical framework to convert it into business decisions. Key gaps include inability to identify valuable customer segments, unclear loyalty drivers, and no data-backed promotional strategy.

Who are our most valuable customers?

What drives loyalty vs. one-time purchases?

Do discounts build loyalty or just attract bargain hunters ?

Which products and seasons drive the most revenue?

Project Methodology

01
Raw Data
Dataset

- 18 original columns
- Consumer shopping data
- 3,900 transactions

02
Python Cleaning
Kaggle Notebook

- Removed
- duplicates & nulls
- Feature engineering: +8 columns
- Standardised text columns

03
MySQL Analysis
MySQL Workbench

- Exported insights to CSV
- 25 analytical queries
- 10 business dimensions

04
Power BI
3-Page Dashboard

- Interactive slicers
- Navigation buttons
- KPI cards + 15+ visuals

05
Insights & Report
This Presentation

- 5 key findings
- 5 strategic recommendations
- Executive summary

Dataset Overview & Feature Engineering

3,900
Total Transactions

26
Columns After Engineering

$233K
Total Revenue

$59.76
Avg Order Value

Original Columns (18)

- Customer ID · Age · Gender
- Item Purchased · Category
- Purchase Amount (USD)
- Location · Size · Color · Season
- Review Rating · Subscription Status
- Shipping Type · Discount Applied
- Promo Code Used · Previous Purchases
- Payment Method · Frequency of Purchases

Engineered Features (+8)

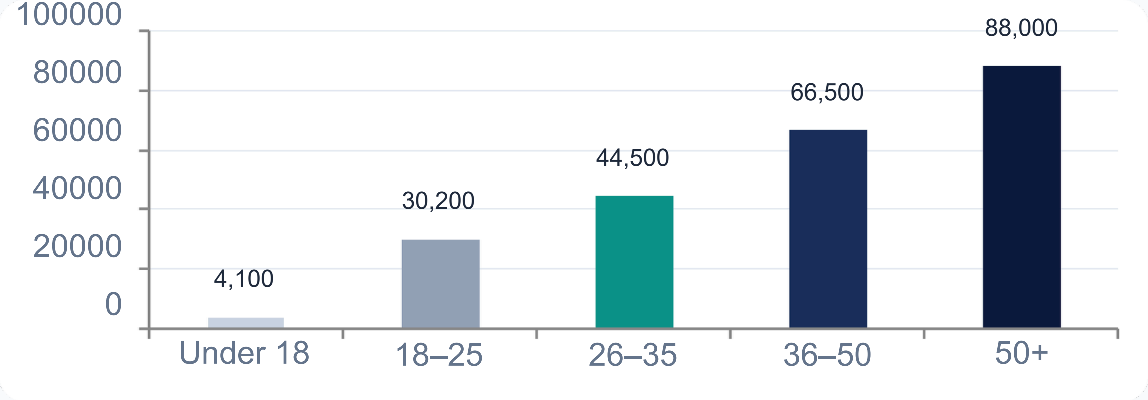
Age Group	Under 18 , 18–25 , 26–35 , 36–50 , 50+
Repeat Buyer	Yes/No flag from Previous Purchases
Purchase Freq. Segment	New / Occasional / Regular / Loyal
Purchase Value Segment	Low / Medium / High (quantile-based)
Any Discount	Combined Discount + Promo flag
Annual Purchase Freq.	Numeric visits per year
Loyalty Score	Frequency + purchase history composite
Est. Historical Spend	Purchase Amt × Previous Purchases

Finding 01 · Customer Segments

Demographics

50+

highest total revenue



Male customers aged 50+ generate the highest revenue AND have the highest loyalty score (49.7)

Younger customers (18–25) purchase frequently but spend significantly less per transaction

Payment preferences correlate with age

Finding 02 · The Footwear Paradox

Product Strategy

More customers ≠ More value per customer

Category	Total Customers	Total Revenue	Avg Order Value	Avg Loyalty Score	Verdict
Clothing	1,737	$104K	$59.9	45.1	High Volume
Foot wear	**974**	**$59K**	**$60.5**	**45.8**	**Higher Value**
Accessories	852	$50K	$58.6	44.9	Stable
Outerwear	337	$20K	$59.3	45.0	Niche

Strategic Implication

The company is over investing in Clothing acquisition while under investing in Footwear the category with the most loyal and individually valuable customers. A targeted Footwear loyalty program would yield higher ROI per customer acquired.

Finding 03 · Discounts Are Not Building Loyalty

Critical Finding

42.1
Discount Users

48.3
Non-Discount Users

14.7% lower loyalty among discount users

Bargain Hunter Pattern

Older male customers (50+) are the most profitable discount users with high loyalty (49.7) but this is despite discounts, not because of them.

Promo Code vs. Discount

Customers using only promo codes show slightly better repeat rates than direct discount recipients, suggesting code-based rewards work marginally better.

Under-18 Discount Trap

The youngest segment uses discounts frequently but contributes the least revenue and lowest loyalty scores ,a clear inefficiency in the current promotional mix.

Finding 04 · Subscription Model Needs Redesign

Retention

Metric	Subscribed	Non-Sub	
Avg Loyalty Score	52.85	42.37	+25% higher
Avg Purchase Amount	$59.8	$59.7	≈ Same
Est. Lifetime Value	$1,485	$1,421	Minimal diff.

What This Means

Subscribers show 25% higher loyaltyscoresbut generate almost identical lifetime value. The subscription program is creating engagement without converting it into incremental spend a classic loyalty revenue disconnect.

Recommended Fix

Add spend-based milestones (e.g. spend $300 - free shipping next month)

Introduce exclusive early access to new collections for subscribe

Track & surface subscriber-specific savings in-app to reinforce value

Finding 05 · Hidden Revenue Opportunities

Opportunity A : Male Gifting Behavior

Men are buying Dresses

"Dress" ranks #3 in revenue for male customers , a category they would not typically buy for themselves. This is a clear signal of gifting behavior.

Recommendation:

Launch a "Gift for Her" section specifically surfaced to male users. Add gift wrapping, personalized notes, and size guides. Men spend more on jewellery than female customers ,this gifting intent is already proven.

Opportunity B : Shirt Satisfaction Risk

High revenue, lowest satisfaction

"Shirt" is the 2nd highest revenue item for women but holds the lowest satisfaction score across all items at 3.48 / 5.0. This is a retention time bomb.

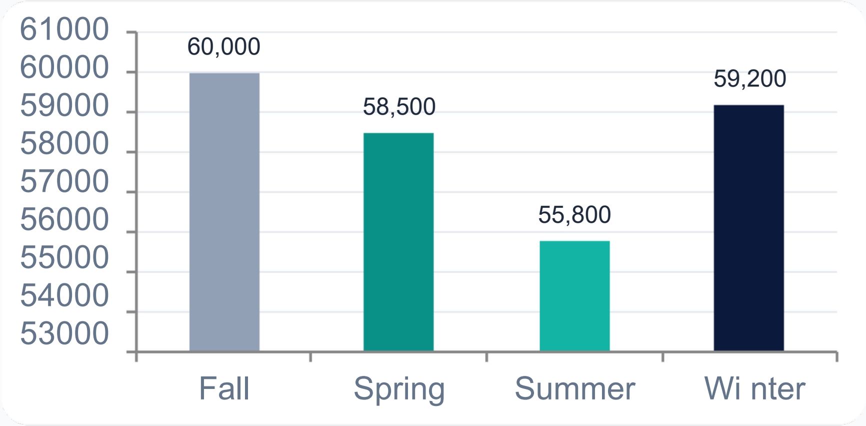
3.48 / 5.0 (Lowest in dataset)

Recommendation:

Conduct a targeted customer feedback campaign on Shirt quality before next season. Investigate sizing accuracy, fabric quality, and return rates. Sunglasses have the highest satisfaction for women(3.92). Study what that product does right and apply it to Shirts.

Finding 06 · Seasonal & Geographic Insights

Revenue by Season



Season ×Category Performance

	Clothing	Footwear	Accessories	O ut erwear
Winter	**High**	Low	S ta ble	Medium
Spring	High	Medium	S ta ble	Low
Summer	Medium	**High**	S ta ble	Low
Fall	High	Medium	S ta ble	Medium

Montana

1 Revenue

Top State

Illinois

Highest Loyalty

Best Retention

Alabama

Highest Repeat Orders

Most Loyal Buyers

Strategic Recommendations

01

High Impact

Target the 50+ Male Segment

This demographic has the highest loyalty, frequency, and spend. Concentrat eat least 30%of paid acquisition budge there.Build look-alike audiences based on this profile.

02

High Impact

Launch a Footwear Loyalty Program

Footwear customers are individually more valuable than Clothing customers. A category-specific program (double points on footwear) will retain high-LTV customers at lower cost.

03

High Impact

Replace Flat Discounts with Points System

Discount users show 14.7% lower loyalty scores. A points-based system rewards repeat behavior rather than one-time bargain seeking ,better ROI on promotional spend.

04

Medium Impact

Redesign Subscription with Spend Milestones

Add spend-triggered rewards (e.g. spend $300 - free shipping nextquarter). Convertthe25%loyaltyadvantagesubscribers already have into actual incremental revenue.

05

Urgent

Investigate Shirt Product Quality Immediately

Second highest revenue item for women with the lowest satisfaction score (3.48) is a retention risk. Audit sizing, returns, and fabric quality before the next season launch.

THANKYOU